# HOSTEL MANAGEMENT SYSTEM

Member 1 — Requirements Lead
Software Engineering Mini-Project — Part 1
*Version 1.1 — Updated Requirements Baseline*
*Status: Finalized for Part 1 requirements baseline.*

## 1.INTRODUCTION

### *PURPOSE*

This document describes the software requirements for Hostel Management System (HMS). It is a system designed to automate and improve core functions of the hostel administration, such as room allotment, fee collection, complaints, visitors' logging, mess administration, leave and outpass, and facility reservation.

### *SCOPE*

This system provides the functionality that enables students to handle their hostel related activities (payment of fee, raising complaints, submitting leave applications, access of mess menu, etc.) through a self-service portal whereas wardens and admin handle room allotments and other record keeping.

### *INTENDED AUDIENCE*

This document is prepared for the development team, project reviewers and course instructors evaluating the mini-project.

### *DEFINITIONS, ACRONYMS AND ABBREVIATIONS*

- HMS : Hostel Management System
- FR : Functional Requirement
- NFR : Non-Functional Requirement
- SO : Security Objective
- SR : Security Requirement
- BR : Business Rule
- Warden: Hostel staff member responsible for approvals and day-to-day supervision

### *REFERENCES*

ISO/IEC/IEEE 29148:2018 Systems and software engineering, Life cycle processes. Requirements engineering

### *OVERVIEW*

Section 2 describes the context of the product. Functional and non-functional requirements are described in Section 3. Use case model is explained in Section 4. Security goals and requirements are discussed in Section 5. Business/validation rules are described in Section 6.

# 2.OVERALL DESCRIPTION

## *PRODUCT PERSPECTIVE*

*The HMS is a fresh, independent system based on the Web with a common database. The interaction between the product and the users (students, wardens, and administrators) takes place via role-based logins.*

## *PRODUCT FUNCTIONS*

- *Room allocation and student record management*
- *Hostel fee tracking and payment recording*
- *Complaint registration and resolution tracking*
- *Visitor entry/exit logging*
- *Mess menu publishing and attendance*
- *Leave/outpass request and approval workflow*
- *Facility booking*
- *Notice broadcast to students*

## *USER CLASSES AND CHARACTERISTICS*

- Student : complains, pays fees, applies for leave, checks mess menu, reserves facilities
- Warden : approves leave applications, maintains visitors' log, deals with complaints
- Administrator : assigns rooms, maintains student database, system configuration

## *OPERATING ENVIRONMENT*

Browser-based application available through regular browsers; backend runs on a relational database server.

## *CONSTRAINTS*

- Should be implemented in the mini-project period
- Role based authentication should be there for all classes of users

## *ASSUMPTIONS AND DEPENDENCIES*

- Students and warden have access to an internet connected device
- Hostel fee amounts and room inventory are predefined by the administrator

# 3.SPECIFIC REQUIREMENTS

## *FUNCTIONAL REQUIREMENTS*

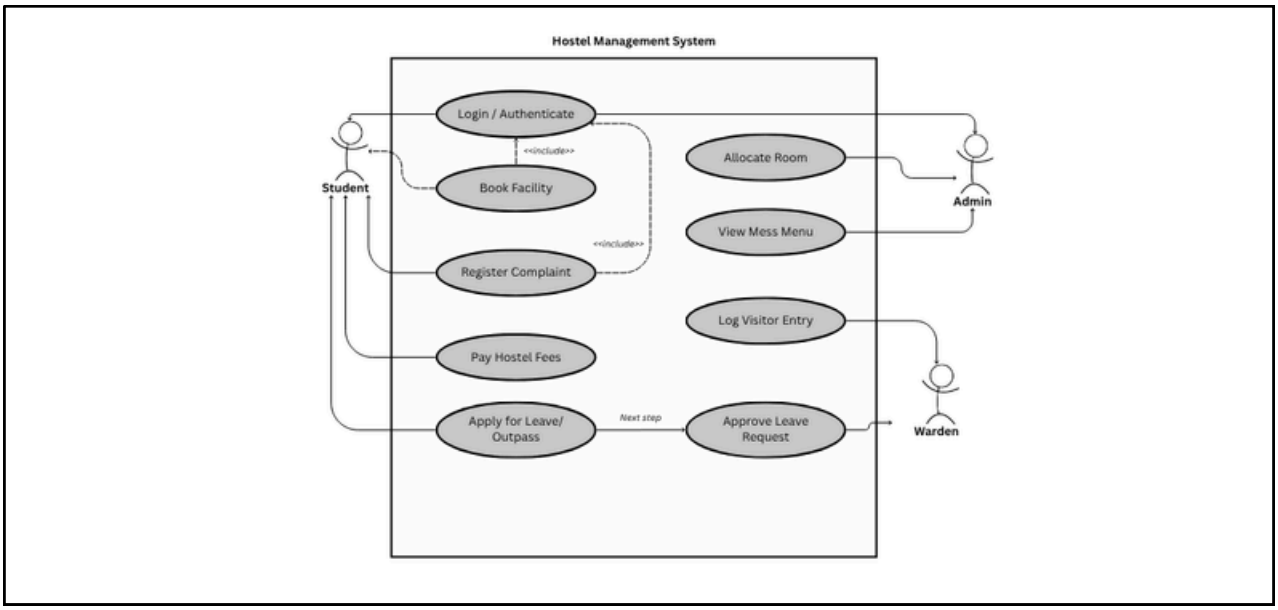
| ID | REQUIREMENT |
|---|---|
| FR - 01 | Admin allocates hostel room to a registered student |
| FR - 02 | Admin shall create and maintain student profile records students may view but not edit their own record |
| FR - 03 | Student views outstanding fees and records fee payments |
| FR - 04 | Student registers complaint; status tracked to resolution |
| FR - 05 | Warden logs visitor entry/exit against a student's room |
| FR - 06 | Admin publishes weekly mess menu visible to all students |
| FR - 07 | Student submits leave/outpass request; warden approves/rejects |
| FR - 08 | Admin broadcasts notices to all students |
| FR - 09 | Student views facility availability and books a facility |

## *NON-FUNCTIONAL REQUIREMENTS*

| ID | REQUIREMENT |
|---|---|
| NFR-01 | System shall respond to a standard user request within 2 seconds under normal load |
| NFR-02 | System shall support at least 500 concurrent student users |
| NFR-03 | System shall be available at least 99% of the time, excluding scheduled maintenance |
| NFR-04 | System interface shall be usable by hostel staff with no prior technical training |

# 4.USE CASE MODEL

## *USE CASE DIAGRAM*




## *USE CASES*

| ID | Use Case | Actor(s) | Relationship |
|---|---|---|---|
| UC-01 | Register Complaint | Student | <<include>> Login |
| UC-02 | Apply for Leave/Outpass | Student | Sequential with UC-08 (not <<extend>>) |
| UC-03 | Login/Authenticate | Student, Warden, Admin | — |
| UC-04 | Allocate Room | Admin | — |
| UC-05 | Log Visitor Entry | Warden | — |
| UC-06 | View Mess Menu | Student | — |
| UC-07 | Pay Hostel Fees | Student | — |
| UC-08 | Approve Leave Request | Warden | Sequential with UC-02 |
| UC-09 | Book Facility | Student | <<include>> Login |

### *USE CASE DESCRIPTIONS (PRIMARY)*

*UC-01: Register Complaint*

- *Actor: Student. Prerequisites: Student is authenticated (Login included). Main Flow: Student submits details about the complaint; complaint is logged and dated; warden gets notification. Postcondition: Complaint is logged with "Open" status.*

*UC-02: Apply for Leave/Outpass*

- *Actor: Student. Prerequisites: Student is authenticated. Main Flow: Student submits dates and reason for leave; request is forwarded to warden for approval (followed by UC-08). Postcondition: Request is in Pending state.*

## 5.SECURITY SECTION

### *SECURITY OBJECTIVES*

| **ID** | **Objective** |
|---|---|
| SO-01 | Protect student personal and payment data from unauthorized access or disclosure. |
| SO-02 | Ensure only authorized staff can modify room allocation and fee records. |

### *SECURITY REQUIREMENTS*

| **ID** | **Requirement** |
|---|---|
| SR-01 | The system shall require authentication (hashed password) for all Admin, Warden, and Student logins. |
| SR-02 | The system shall enforce role-based access control distinguishing Student, Warden, and Admin permissions. |

## 6.BUSINESS/VALIDATION RULES

| ID | Rule |
|---|---|
| BR-01 | A student cannot be allocated more than one active room at a time. |
| BR-02 | Leave/outpass requests require warden approval before being marked "Approved". |
| BR-03 | Partial fee payments are permitted; the fee record shall only be marked "Paid" when the outstanding balance reaches zero. |
| BR-04 | Complaint status must follow Open → In Progress → Resolved (no skipping). |
| BR-05 | Visitor log entry requires both entry and exit timestamp fields (exit may be null while visitor is on-premises). |
| BR-06 | A facility booking cannot be made if the facility is already booked for that time slot. |

# 7.REQUIREMENTS TRACEABILITY MATRIX

| Req ID | Description | Traced To |
|---|---|---|
| FR-01 | Room allocation | Test Plan TC-003 / Architecture |
| FR-02 | Student records (Admin-owned, Student read-only) | Test Plan TC-004 / Architecture |
| FR-03 | Fee payment | Test Plan TC-005 / Architecture |
| FR-04 | Complaint registration | Test Plan TC-006 / Architecture |
| FR-05 | Visitor logging | Test Plan TC-007 / Architecture |
| FR-06 | Mess menu | Test Plan TC-008 / Architecture |
| FR-07 | Leave/outpass | Test Plan TC-009 / Architecture |
| FR-08 | Notice broadcast | Test Plan TC-010 / Architecture |
| FR-09 | Facility booking | Test Plan TC-011 / Architecture |
| NFR-01 to NFR-04 | Performance, load, availability, usability | Test Plan TC-012 to TC-015 |
| SO-01, SO-02, SR-01, SR-02 | Security objectives and requirements | Test Plan Section 5.1 / Security Architecture |
| BR-01 to BR-06 | Business/validation rules | Test Plan TC-003, TC-005, TC-006, TC-007, TC-009, TC-011 |